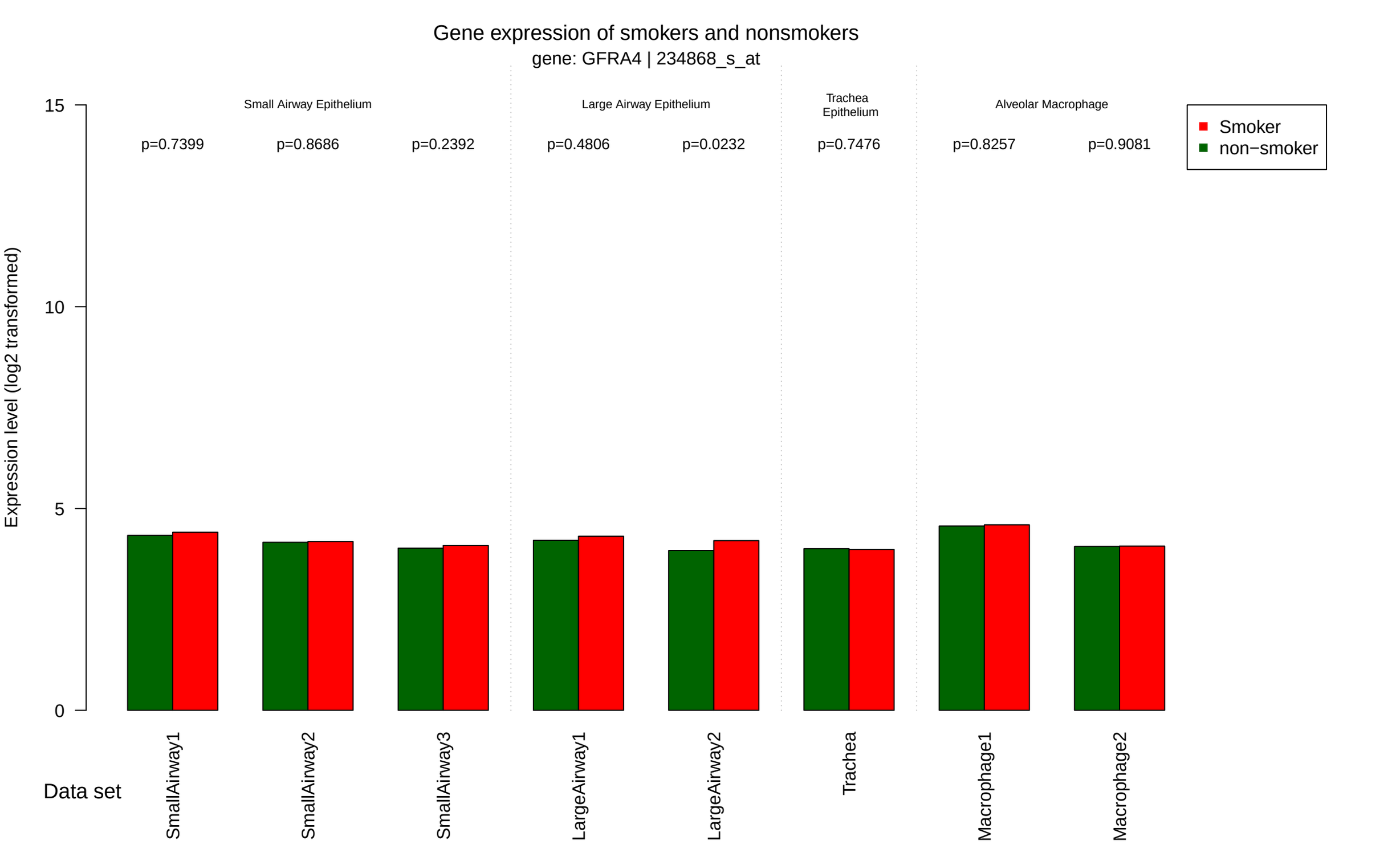
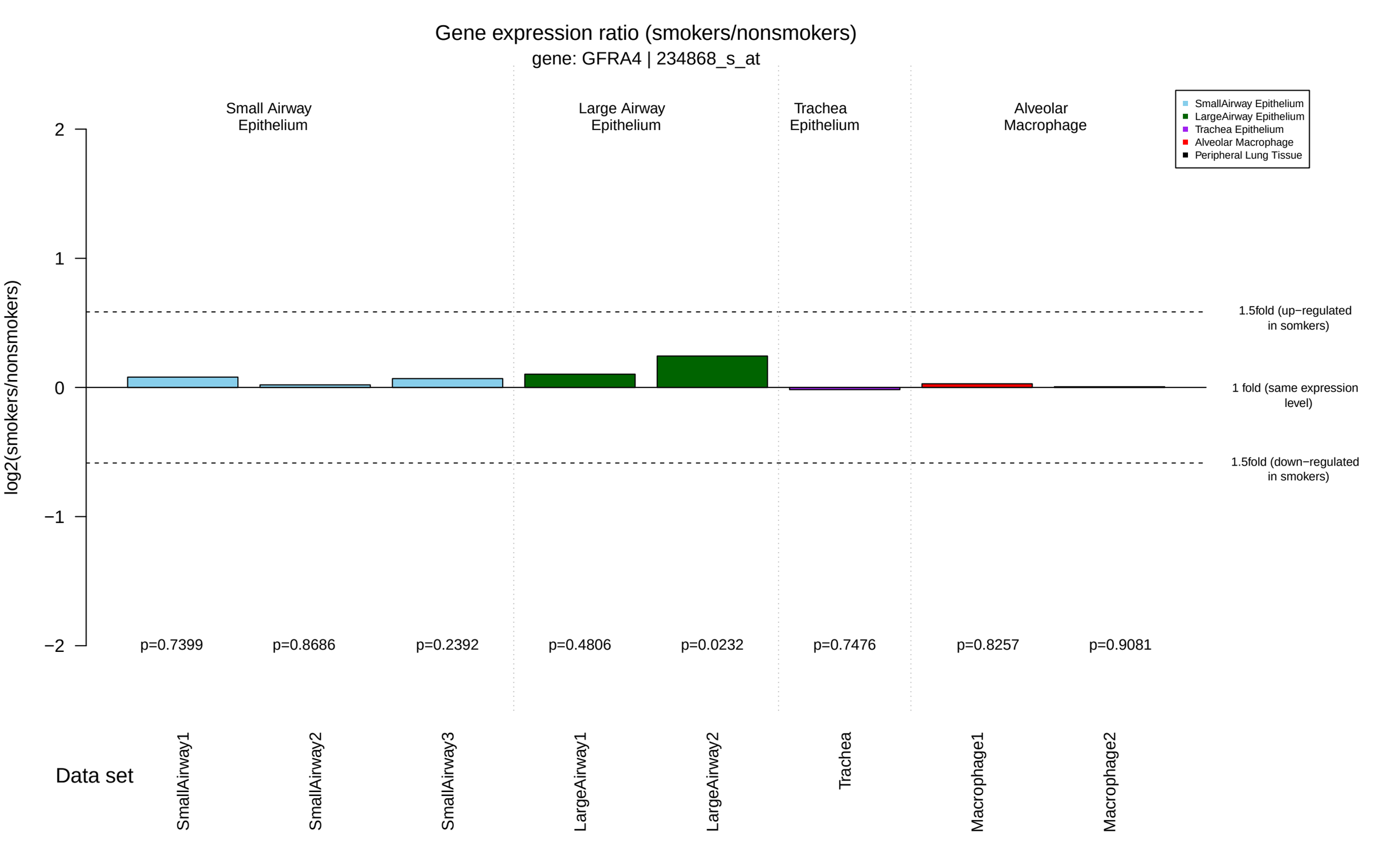
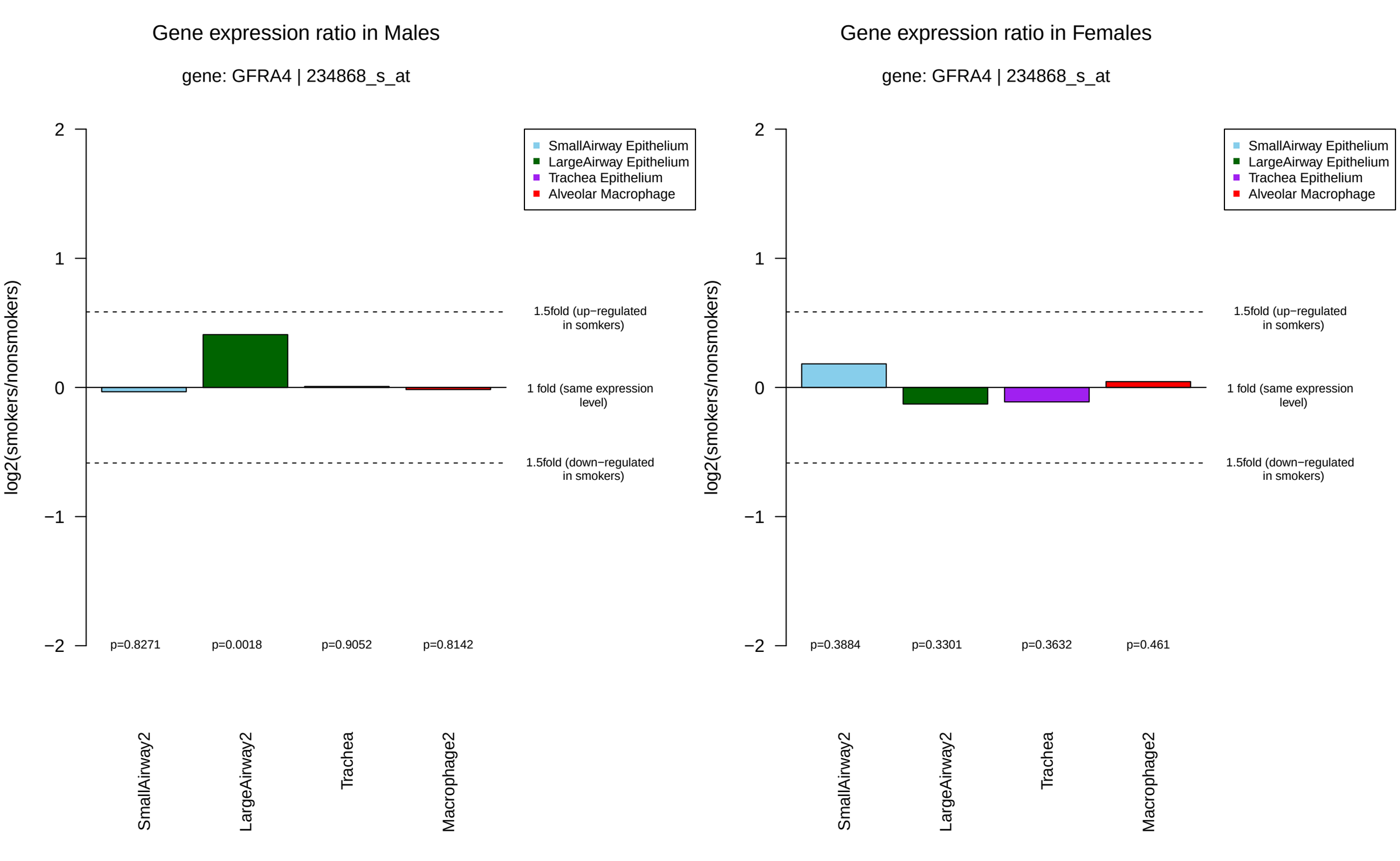
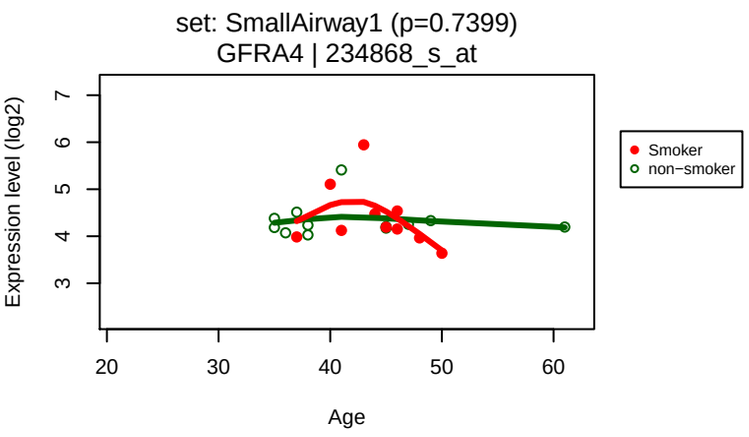
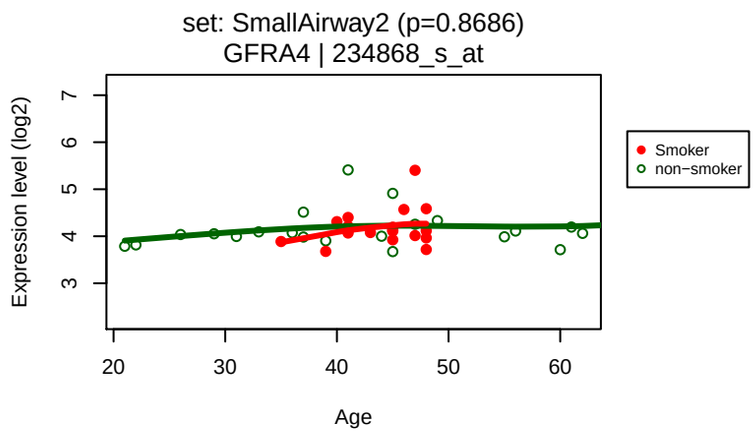
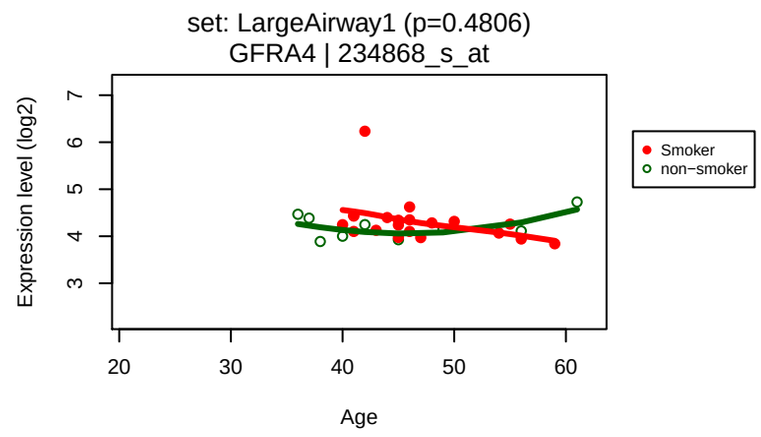

Gene expression of smokers and nonsmokers
gene: GFRA4 | 234868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7399
p=0.8686
p=0.2392
p=0.4806
p=0.0232
p=0.7476
p=0.8257
p=0.9081
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

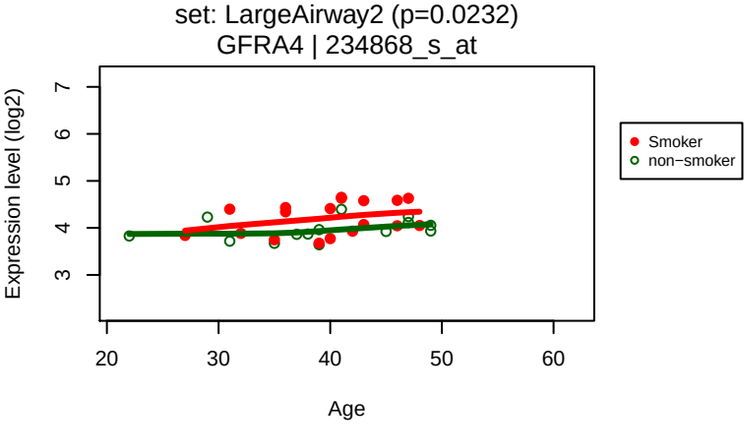

Gene expression ratio (smokers/nonsmokers)
gene: GFRA4 | 234868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7399
p=0.8686
p=0.2392
p=0.4806
p=0.0232
p=0.7476
p=0.8257
p=0.9081
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

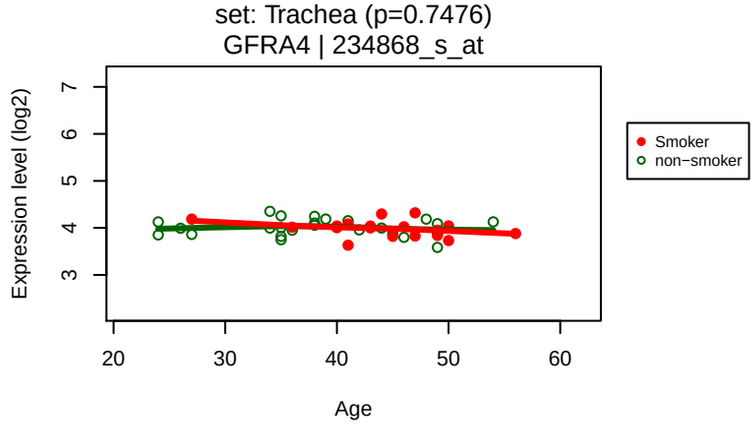

Gene expression ratio in Males
gene: GFRA4 | 234868_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8271
p=0.0018
p=0.9052
p=0.8142
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GFRA4 | 234868_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3884
p=0.3301
p=0.3632
p=0.461
SmallAirway2
LargeAirway2
Trachea
Macrophage2

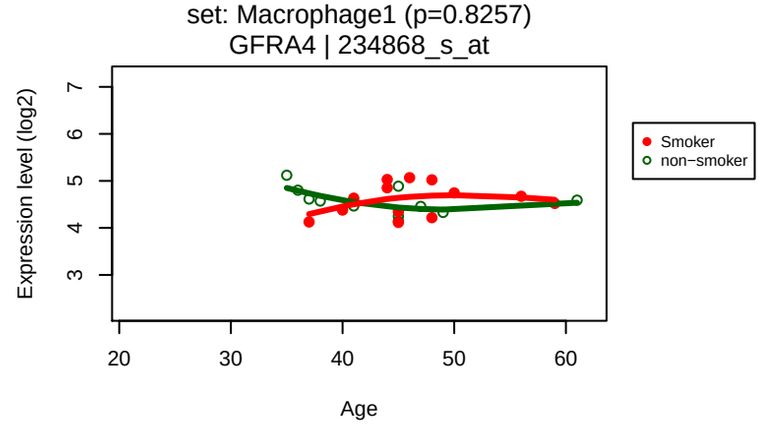

set: SmallAirway1 (p=0.7399)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker

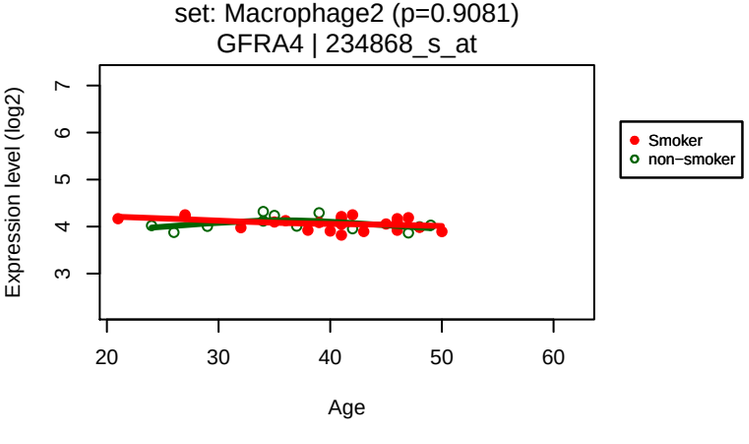

set: SmallAirway2 (p=0.8686)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker

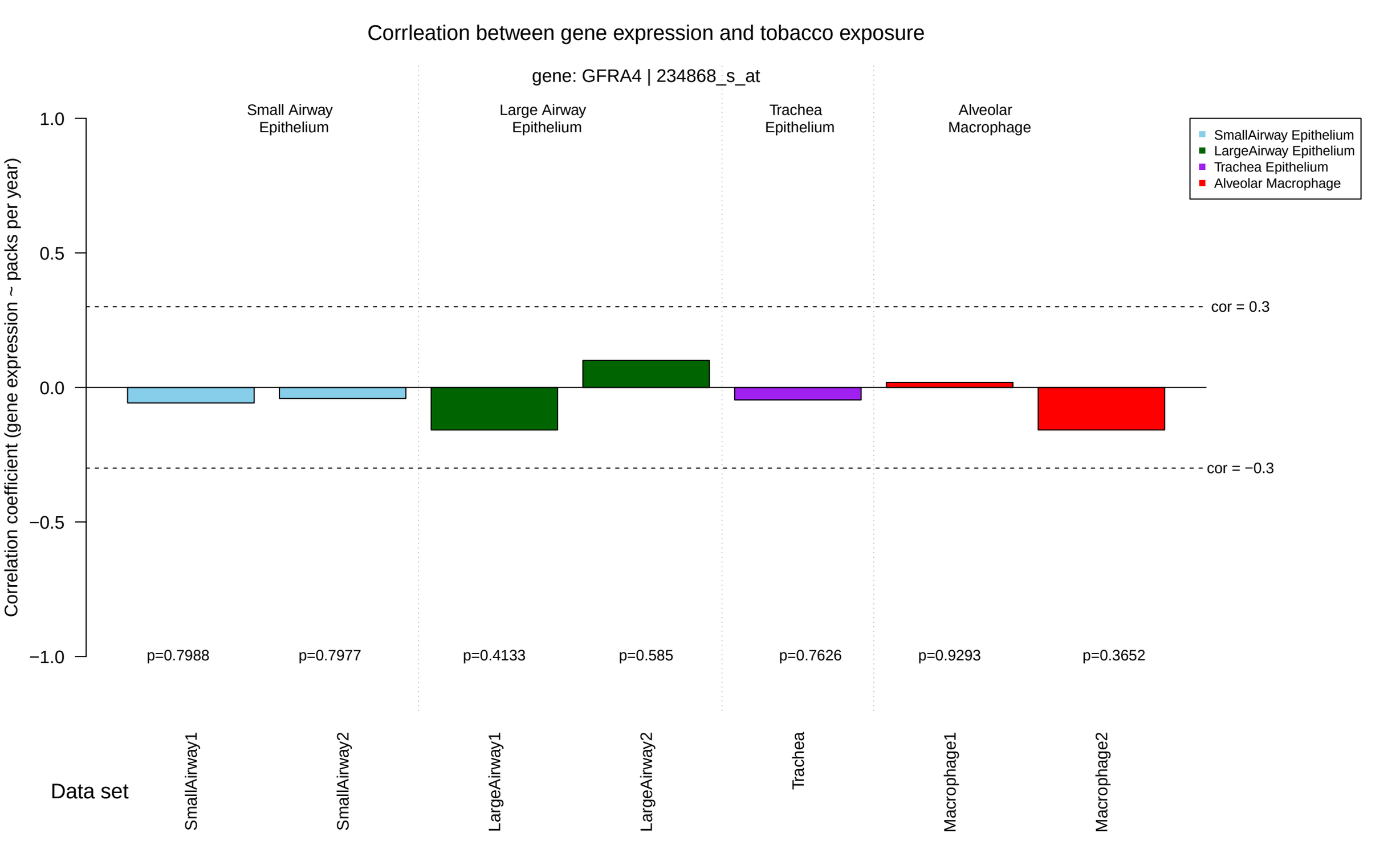

set: LargeAirway1 (p=0.4806)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0232)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7476)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8257)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9081)
GFRA4 | 234868_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GFRA4 | 234868_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7988
p=0.7977
p=0.4133
p=0.585
p=0.7626
p=0.9293
p=0.3652
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2